A THREE-QUARTER EXCEPTIONAL-SET BOUND FOR THE ERDŐS–STRAUS EQUATION

ANONYMOUS

Abstract. Let $E(N)$ be the number of $n \le N$ for which $4/n$ is not a sum of three unit fractions. Vaughan (1970) proved $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$. We prove

$$E(N) \ll N \exp\{-c(\log N)^{3/4}\},$$

with an absolute, ineffective constant $c > 0$. We are not aware of an earlier improvement of the exponent $2/3$. The forced residue classes are those of the classical Type II identity, but a prime $\ell \le X$ is used together with a *multiplier* $k \le X^{\kappa}$, giving classes modulo $k\ell$. Conditionally on n mod $\mathrm{lcm}(k)$ the hits at different primes are independent, and Bombieri–Vinogradov, Shiu's theorem and a Cauchy–Schwarz argument show that every conditional fibre has expected number of hits $\asymp (\log X)^3$, against $(\log X)^2$ for Vaughan's classes. Instead of the large sieve we count on $[1, N]$ an explicit nonnegative majorant made of congruence classes, namely a small-prime selector times an even Bonferroni truncation. Its coefficient ledger costs $\exp\{O((\log X)^4)\}$, and balancing gives the exponent $3/4$. This note has been checked internally only; it is not a proof of the Erdős–Straus conjecture.

1. Introduction

Call a positive integer n *representable* if there are positive integers x_1, x_2, x_3, not necessarily distinct, with

$$\frac{4}{n} = \frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3}. \tag{1}$$

The Erdős–Straus conjecture asserts that every $n \ge 2$ is representable. Put

$$E(N) = \#\{1 \le n \le N : n \text{ not representable}\},$$
$$E_{\mathrm{pr}}(N) = \#\{p \le N \text{ prime} : p \text{ not representable}\}.$$

(Here $n = 1$ is not representable; this is irrelevant for a density bound.)

Theorem 1.1. *There are absolute constants $c_*, C_* > 0$ such that for every real $N \ge 3$*

$$E_{\mathrm{pr}}(N) \le E(N) \le C_* N \exp\{-c_*(\log N)^{3/4}\}.$$

The constants are ineffective, because the proof uses the Bombieri–Vinogradov theorem.

Background. Vaughan [V] proved $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$. This is still the bound quoted by Elsholtz and Tao [ET, §1] and in the survey of Bloom and Elsholtz [BE]. Pomerance and Weingartner [PW, Thm. 1.3] give a version that is uniform in the numerator m: for $4 \le m \le (\log N)^2$, at most $N \exp\{-C(\log^2 N/\varphi(m))^{1/3}\}$ integers $n \le N$ have m/n not a sum of three unit fractions. For fixed m the exponent

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is again 2/3. For sums of $k \geq 3$ unit fractions, Elsholtz's bound has exponent $1 - 1/(2^{k-1} - 1)$ [El], [BE, §on exceptional sets]. This is 2/3 for $k = 3$; larger k is a different problem. A companion internal note [LL] obtains the intermediate bound $N \exp\{-c(\log N)^{2/3}(\log\log N)^{1/3}\}$ by the large sieve with multipliers $k \ll \log N$. A search of the literature up to 2026 found no improvement of the exponent 2/3 for three unit fractions. This is a report of a search, not a priority claim.

Where the exponents come from. For a prime $\ell \equiv 3 \pmod 4$, every divisor pair of $(\ell+1)/4$ forces a residue class modulo ℓ to consist of representable integers. On average over primes $\ell \leq X$ the mass of these classes is $\sum_\ell \omega(\ell)/\ell \asymp (\log X)^2$. To turn mass μ at primes $\leq X$ into a saving $\mathrm{e}^{-c\mu}$, any sieve of this kind must control intersections of about μ classes. Such intersections have moduli of size up to $X^{O(\mu)}$, and this costs $\exp\{O(\mu \log X)\}$ in the error terms. With $t = \log X$ and $\mu \asymp t^{\mathcal{B}}$, the balance $t^{\mathcal{B}+1} \asymp \log N$ gives a saving $(\log N)^{\mathcal{B}/(\mathcal{B}+1)}$. Vaughan's large sieve has $\mathcal{B} = 2$, hence the exponent 2/3.

The point of this note is $\mathcal{B} = 3$. For a multiplier $k \equiv 1 \pmod 4$ up to $K = X^\kappa$, factorizations of $(k\ell+1)/4$ force classes modulo $k\ell$ (Lemma 2.1). Conditionally on $n \bmod k$, such a class is a class modulo ℓ. Summed over k, the extra mass is the factor $\sum_k \varphi(k)/k^2 \asymp \log K \asymp t$ (Theorem 4.2). There is a difficulty: $L_K = \mathrm{lcm}\{k \leq K\}$ satisfies $\log L_K \asymp K$, so L_K is far larger than N, so one cannot split $[1, N]$ into progressions modulo L_K as [LL] does. We avoid any such splitting. The majorant of Section 7 is a fixed finite signed combination of congruence classes. Its mean is computed exactly in a finite CRT probability space. The fibres modulo L_K appear only inside that computation, and its count on $[1, N]$ costs one unit per term. The terms number $\exp\{O(t^4)\}$, so $t \asymp (\log N)^{1/4}$ and the saving is $t^3 \asymp (\log N)^{3/4}$.

Remark 1.2 (the method limit). Internal, unrefereed companion work [KA3, SL] shows that 3/4 is optimal for the architecture used here. More precisely, consider any nonnegative CRT majorant that is ≥ 1 on all integers avoiding a finite family of forced classes (the classes $\mathcal{R}(M) = \{-4D \bmod M : D \mid ((M+1)/4)^2\}$ of Remark 9.2, and the (a, D) and Case A classes in the notation of [KA3]) and of selector classes $0 \bmod p$, and that is used through $N\mathbb{E}\nu + \sum |\text{coefficients}|$ with coefficient sum $< N$. Every such majorant saves at most $C(\log N)^{3/4}$; see [KA3, Cor. 4.2] and [SL, Thm. 10.14]. The majorant of Theorem 7.2 is of this form. These results do not cover large sieves of super-polynomial level, hybrid interval methods, or non-CRT inputs. Nothing in this remark is used below.

Status. The proof below has been checked by internal audits only. It has not been refereed or read by an external expert. An earlier version of the argument proved the supply estimate of Section 4 by an incidence-congestion pruning with a cutoff on $\omega(uv)$, and gave a second proof of Theorem 5.1 by an ordered-atom induction. Both are superseded by the shorter arguments here and have been removed.

Parameters. Fix $\kappa = 1/480$; any fixed $0 < \kappa < 1/240$ would do. Throughout,

$$\begin{gathered} t = \log X, \quad K = \lfloor X^\kappa \rfloor, \quad H = K^{10}, \\ \mathcal{K}(K) = \{1 \leq k \leq K : k \equiv 1 \pmod 4\}, \quad L_K = \mathrm{lcm}_{k \in \mathcal{K}(K)}\, k. \end{gathered} \tag{2}$$

The constants are chosen in the following order. First come the absolute analytic constants (Shiu, Brun–Titchmarsh, Bombieri–Vinogradov with a fixed exponent) and the lattice threshold K_0. Then the void constant c_v is chosen (Section 6), then

the selector constant B in $y = Bt^3$, then the Bonferroni constant D_B, then the ledger constant C_0, and finally α in $t = \alpha(\log N)^{1/4}$. Every later "sufficiently large" threshold depends only on the constants already chosen, and none depends on a residue class, a subfamily of multipliers, a moment order, or N. We write $A \ll B$ with constants that are absolute in this sense.

The guiding distinction is between using prime distribution at scale X to *construct* classes and counting those classes on *all* integers up to N. No prime-counting estimate is ever used at scale N. Integer counts in residue classes, each with error at most one, take its place.

2. Forced classes and a fixed atom family

Lemma 2.1 (multiplier identity)**.** *Let k, ℓ be positive integers with $k\ell \equiv 3 \pmod 4$, and let $(k\ell+1)/4 = uvw$ with positive integers u, v, w. If $n \geq 1$ satisfies $nv \equiv -u \pmod{k\ell}$, then $s = (nv+u)/(k\ell)$ is a positive integer and*

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw}.$$

Thus every positive integer in the class $-uv^{-1} \pmod{k\ell}$ is representable. The inverse exists because uvw is coprime to $k\ell$.

Proof. Over the common denominator $nsuvw$ the numerators add up to $nv+u+s = s(k\ell+1) = 4suvw$. □

For $k = 1$ these are the classes used by Vaughan; the solvability criterion is classical in the form "$4abd - 1 \mid b + nc$" (see [ET, Prop. 2.7], where it is attributed to [V]). The multiplier k only changes which moduli $M = k\ell$ are used.

Blocks and atoms. Put $x_j = 2^j X^{1/2}$ and use the disjoint blocks

$$I_j = (x_j, \min(2x_j, X)], \qquad 0 \leq j < \left\lceil \frac{\log X}{2\log 2} \right\rceil,$$

with $z_j = x_j^{1/6}$. A block is *full* if $2x_j \leq X$. Lower estimates use only full blocks, and upper estimates enlarge the last block to a full one. There are $\asymp t$ blocks, and $t/2 \leq \log x_j \leq t$. Since $\kappa < 1/240$, for all sufficiently large X every block satisfies

$$z_j \geq X^{1/12} > K^{20} = H^2, \qquad \Lambda_j := \log(z_j/H) \geq \tfrac{1}{2}\log z_j. \tag{3}$$

Let $\mathcal{A}_X$ be the set of quadruples $A = (k, \ell, u, v)$ with

$$\begin{gathered} k \in \mathcal{K}(K), \quad \ell \in I_j \text{ prime for some } j, \quad H < u, v \leq z_j, \\ (u, v) = (uv, k) = 1, \quad 4uv \mid k\ell + 1. \end{gathered} \tag{4}$$

Divisibility forces $\ell \equiv 3 \pmod 4$. To each atom A attach the class

$$E_A = \{n : n \equiv -uv^{-1} \pmod{k\ell}\}, \qquad H_X(n) = \sum_{A \in \mathcal{A}_X} \mathbf{1}_{E_A}(n). \tag{5}$$

By Lemma 2.1 with $w = (k\ell+1)/(4uv)$, a non-representable n has $H_X(n) = 0$. The family $\mathcal{A}_X$ is fixed: it does not depend on N or on any residue class.

The probability space. Let $2 \le y < K$ and $P_y = \prod_{p \le y} p$. We use the uniform measure on $\mathbb{Z}/\mathcal{M}\mathbb{Z}$, where

$$\mathcal{M} = \operatorname{lcm}(L_K, P_y) \prod_{\substack{X^{1/2} < \ell \le X \\ \ell \text{ prime}}} \ell. \tag{6}$$

Every class $a \bmod q$ with $q \mid \mathcal{M}$ has probability $1/q$. The size of $\mathcal{M}$ is irrelevant: it only defines probabilities, and $[1, N]$ is never partitioned modulo $\mathcal{M}$. We write $c = n \bmod L_K$ and call $\{n : n \equiv c\ (L_K)\}$ the *fibre* of c.

Lemma 2.2 (distinctness and conditional independence)**.** *For all sufficiently large X the following hold.*

1. *Distinct atoms with the same prime ℓ have distinct classes modulo ℓ.*
2. *Conditionally on $c = n \bmod L_K$, and on any event determined by $n \bmod P_y$, the residues $n \bmod \ell$ for the primes $X^{1/2} < \ell \le X$ are independent and uniform.*
3. *An atom A meets the fibre of c if and only if $k \mid u + cv$. In that case $E_A \cap \{n \equiv c\ (L_K)\} = \{n \equiv c\ (L_K),\ n \equiv -uv^{-1}\ (\ell)\}$.*

Consequently, in the fibre of c, $H_X = \sum_\ell I_\ell$ is a sum of independent Bernoulli variables, with $\mathbb{P}(I_\ell = 1 \mid c) = f_c(\ell)/\ell$, where

$$f_c(\ell) = \#\{A \in \mathcal{A}_X : \ell_A = \ell,\ k_A \mid u_A + cv_A\}. \tag{7}$$

Proof. (1) Let (k, ℓ, u, v) and (k', ℓ, u', v') have the same class modulo ℓ, with $\ell \in I_j$. Then $\ell \mid uv' - u'v$ and $|uv' - u'v| < z_j^2 < x_j < \ell$, so $uv' = u'v$. Both fractions are reduced, so $(u, v) = (u', v')$. Next $k\ell \equiv k'\ell \equiv -1 \pmod{4uv}$ and $(\ell, 4uv) = 1$ give $k \equiv k' \pmod{4uv}$. Since $4uv > 4H^2 > K$, $k = k'$. (2) The primes $\ell > X^{1/2} > K > y$ are coprime to $\operatorname{lcm}(L_K, P_y)$, so this is the Chinese remainder theorem. (3) Since $k \mid L_K$, the condition $n \equiv -uv^{-1} \pmod k$ is determined by c, and it holds if and only if $k \mid u + cv$. By (1) the active atoms at ℓ occupy $f_c(\ell)$ distinct classes modulo ℓ, and at most one of them contains n. □

Note that $H_X(n) = \sum_\ell I_\ell(n)$ holds identically and not just in distribution: at each ℓ at most one atom can contain n.

3. Analytic inputs and lattice sums

Prime distribution. We use Brun–Titchmarsh in the form of Montgomery and Vaughan [MV, Thm. 2]: for $1 \le q < Y$ and $(a, q) = 1$,

$$\pi(x + Y; q, a) - \pi(x; q, a) \le \frac{2Y}{\varphi(q) \log(Y/q)}. \tag{8}$$

With $Y = x$ and $q \le 4x^{1/3}$ the right side is at most $4x/(\varphi(q) \log x)$ for large x.

For $(a, q) = 1$ put

$$E_x(q, a) = \pi(2x; q, a) - \pi(x; q, a) - \frac{\operatorname{li}(2x) - \operatorname{li}(x)}{\varphi(q)}, \qquad E_x^*(q) = \max_{(a,q)=1} |E_x(q, a)|.$$

By the Bombieri–Vinogradov theorem, for every fixed $R > 0$ there are A_R, C_R, x_R such that for $x \ge x_R$

$$\sum_{q \le x^{1/2} (\log x)^{-A_R}} E_x^*(q) \le C_R x (\log x)^{-R}. \tag{9}$$

Davenport [D, Ch. 28] states the theorem for $\max_{y\le x}\max_a|\psi(y;q,a)-y/\varphi(q)|$. The form with π and li follows by partial summation, since prime powers contribute negligibly, and taking two endpoints costs a factor 2. The constants are ineffective. Only moduli $q\le 4x^{1/3}$ occur below, and they lie below the level $x^{1/2}(\log x)^{-A_R}$ for large x.

Shiu's theorem. Let $F\ge 0$ be multiplicative with $F(p^a)\le A_1^a$ for all prime powers, and $F(n)\le A_2(\epsilon)n^\epsilon$ for every $\epsilon>0$. Fix $0<\alpha_s,\beta_s<1/2$. Shiu [S, Thm. 1] proves that

$$\sum_{\substack{V-Y<n\le V\\ n\equiv a\ (q)}} F(n)\ll \frac{Y}{\varphi(q)\log V}\exp\Big\{\sum_{p\le V,\ p\nmid q}\frac{F(p)}{p}\Big\}, \tag{10}$$

uniformly for $0<a<q$, $(a,q)=1$, $q<Y^{1-\alpha_s}$ and $V^{\beta_s}<Y\le V$. The implied constant depends only on α_s,β_s,A_1,A_2. We use it only for

$$F(n)=2^{\omega(n)}\frac{n}{\varphi(n)},\qquad F(p^a)=\frac{2p}{p-1}\le 4,\qquad F(n)\le 4^{\omega(n)}\le\tau(n)^2\ll_\epsilon n^\epsilon,$$

in *boxes* $(U,\frac{21}{20}U]$ with $U\ge H=K^{10}$ and moduli $q\le K^2$. We take $V=\frac{21}{20}U$, $Y=\frac{1}{20}U$ and $\alpha_s=\beta_s=\frac14$. Then $q\le K^2\le U^{1/5}<Y^{3/4}$ and $V^{1/4}<Y$ as soon as K is large. An interval $(H,z]$ is covered by consecutive boxes starting at $U=H$, and the last box is enlarged. This uses $O(\log z)$ boxes, all with endpoints $\le\frac{21}{20}z$. The modulus $q=1$ is excluded in [S]. For it we use instead the elementary bound $\sum_{n\le V}F(n)\ll V\log V$, which follows from $F=1*h$, where $h\ge0$ is supported on squarefree integers and $h(p)=(p+1)/(p-1)$. Indeed $\sum_{n\le V}F(n)\le V\sum_{d\le V}h(d)/d\ll V\log V$.

Lattice sums. For $k\ge1$, $(c,k)=1$ and $1\le H<z$, put $\Lambda=\log(z/H)$ and

$$\mathcal{S}(k,c;H,z)=\{(u,v):H<u,v\le z,\ (u,v)=1,\ (v,k)=1,\ u\equiv -cv\ (\mathrm{mod}\ k)\}.$$

Then $(u,k)=1$ automatically.

Lemma 3.1 (lattice sums; cf. [LL, Lemma 3.1])**.** *There are absolute K_0,C_1 such that for every $K\ge K_0$, every $1\le k\le K$, every $(c,k)=1$ and every $z\ge H^2$, where $H=K^{10}$,*

$$\sum_{\mathcal{S}(k,c;H,z)}\frac{1}{uv}\ge\frac14\frac{\varphi(k)}{k^2}\Lambda^2, \tag{11}$$

$$\sum_{\mathcal{S}(k,c;H,z)}\frac{1}{\varphi(u)\varphi(v)}\le C_1\frac{\varphi(k)}{k^2}\Lambda^2. \tag{12}$$

Proof. For $H'\ge1$, $z'>H'$, $\Lambda'=\log(z'/H')$ and $(a,k)=1$, comparison with integrals and Möbius inversion give

$$\sum_{\substack{H'<u\le z'\\ u\equiv a\ (k)}}\frac1u=\frac{\Lambda'}{k}+O\Big(\frac{1}{H'}\Big),\qquad \sum_{\substack{H'<v\le z'\\ (v,k)=1}}\frac1v=\frac{\varphi(k)}{k}\Lambda'+O\Big(\frac{\tau(k)}{H'}\Big), \tag{13}$$

including empty progressions. Hence the double sum $T(H',z')$ over the conditions of $\mathcal{S}$ without $(u,v)=1$ is $\frac{\varphi(k)}{k^2}\Lambda'^2+O(\Lambda'\tau(k)/H')$ for $\Lambda'\ge1$. Remove $(u,v)=1$ by Möbius inversion. A common divisor d is coprime to k, and $(u,v)=(du',dv')$

ranges over the same conditions with $(H/d, z/d]$. For $d \le H$ use the formula for T. For $d > H$ use $T(H/d, z/d) \le (1+\log z)^2$ and $\log z \le 2\Lambda$. This gives

$$\sum_{\mathcal{S}} \frac{1}{uv} = \frac{\varphi(k)}{k^2}\Lambda^2 \sum_{\substack{d \le H \\ (d,k)=1}} \frac{\mu(d)}{d^2} + O\Big(\frac{\Lambda\tau(k)\log z}{H} + \frac{\Lambda^2}{H}\Big).$$

The coefficient is $\prod_{p\nmid k}(1-p^{-2}) + O(H^{-1}) \ge 6/\pi^2 - O(H^{-1})$. Because $k/\varphi(k) \le 2^{\omega(k)} \le \tau(k) \le 2\sqrt{k}$, the error relative to $\varphi(k)\Lambda^2/k^2$ is $O(k\tau(k)^2/H) = O(K^2/H)$. This proves (11) for $K \ge K_0$.

For (12) drop $(u,v)=1$ and write $1/\varphi(m) = m^{-1}\sum_{d\mid m}\mu^2(d)/\varphi(d)$. If $d \mid u$ and $u \equiv a \pmod k$ with $(a,k)=1$, then $(d,k)=1$. By (13) for $d \le H$, and the bound $\sum_{m \le z/d} 1/m \le 1+\log z$ for $d > H$,

$$\begin{aligned}\sum_{\substack{H < u \le z \\ u \equiv a \ (k)}} \frac{1}{\varphi(u)} &\ll \frac{\Lambda}{k}\sum_{d}\frac{\mu^2(d)}{d\varphi(d)} + \frac{1}{H}\sum_{d\le H}\frac{\mu^2(d)}{\varphi(d)} + (1+\log z)\sum_{d>H}\frac{1}{d\varphi(d)} \\ &\ll \frac{\Lambda}{k} + \frac{\log H}{H} + \frac{\Lambda}{\sqrt{H}},\end{aligned}$$

using $\varphi(d) \gg \sqrt{d}$. In the same way $\sum_{H<v\le z,(v,k)=1} 1/\varphi(v) \ll \frac{\varphi(k)}{k}\Lambda + \tau(k)\log H/H + \Lambda/\sqrt{H}$. Since $\varphi(k)/k \ge (2k)^{-1/2}$, $\tau(k) \le 2\sqrt{k}$ and $\Lambda \ge 10\log K$, the secondary terms are dominated by the main terms. Multiply the two bounds, applying the first with $a \equiv -cv$. □

Lemma 3.2 (multiplier harmonic sum; cf. [LL, Lemma 3.2])**.** *For every subset $\mathcal{J} \subseteq \mathcal{K}(K)$ put $h(\mathcal{J}) = \sum_{k\in\mathcal{J}} \varphi(k)/k^2$. For $K \ge 5$,*

$$h(\mathcal{K}(K)) = \frac{2}{\pi^2}\log K + O(1), \tag{14}$$

and for every prime p,

$$\sum_{\substack{k \le K \\ p\mid k}} \frac{\varphi(k)}{k^2} \le \frac{1+\log K}{p}. \tag{15}$$

Proof. From $\varphi(k)/k^2 = k^{-1}\sum_{d\mid k}\mu(d)/d$ and $k \equiv 1 \pmod 4$,

$$h(\mathcal{K}(K)) = \sum_{\substack{d \le K \\ d \text{ odd}}} \frac{\mu(d)}{d^2}\Big(\frac{1}{4}\log\frac{K}{d} + O(1)\Big),$$

and $\sum_{d \text{ odd}} \mu(d)/d^2 = 8/\pi^2$. For (15) use $\varphi(k)/k^2 \le 1/k$ and write $k = pm$. □

4. Cubic supply in every fibre

For $\mathcal{J} \subseteq \mathcal{K}(K)$ let $L_{\mathcal{J}} = \operatorname{lcm}_{k\in\mathcal{J}} k$. For $(c, L_{\mathcal{J}}) = 1$ and integers u, v put

$$r_{\mathcal{J}}(u,v;c) = \#\{k \in \mathcal{J} : k \mid u + cv\}.$$

If $(u,v)=1$ and $k \mid u+cv$ with $(c,k)=1$, then $(uv,k)=1$, so $(u,v) \in \mathcal{S}(k,c;H,z)$ whenever $H < u, v \le z$.

Lemma 4.1 (weighted second incidence moment)**.** *There is an absolute C_2 such that for $K \ge K_0$, $H = K^{10}$, $z \ge H^2$, every $\mathcal{J} \subseteq \mathcal{K}(K)$ and every $(c, L_{\mathcal{J}}) = 1$,*

$$\sum_{\substack{H<u,v\le z\\(u,v)=1}} \frac{2^{\omega(uv)}\, r_{\mathcal{J}}(u,v;c)^2}{\varphi(4uv)} \le C_2 (\log z)^4 (1+\log K)^3. \tag{16}$$

Proof. For coprime u, v we have $\varphi(4uv) \ge 2\varphi(u)\varphi(v)$. This is an equality if uv is odd. If $u = 2^a m$ with $a \ge 1$ and m odd, then $\varphi(4u) = 2^{a+1}\varphi(m) = 4\varphi(u)$. Expanding the square, a pair $k, k' \in \mathcal{J}$ contributes through the single condition $d = [k,k'] \mid u + cv$, where $d \le K^2$ and $(c,d) = 1$. For coprime u, v this condition forces $(v,d) = 1$. We may therefore drop $(u,v) = 1$ but keep $(v,d) = 1$, and then the residue $u \equiv -cv \pmod d$ is reduced. Put $F(n) = 2^{\omega(n)} n/\varphi(n)$, so that $2^{\omega(n)}/\varphi(n) = F(n)/n$. The pair contributes at most

$$\frac{1}{2} \sum_{\substack{H<v\le z\\(v,d)=1}} \frac{F(v)}{v} \sum_{\substack{H<u\le z\\ u\equiv -cv \ (d)}} \frac{F(u)}{u}.$$

Now cover $(H, z]$ by boxes and take $d \ge 2$. In a box $(U, \frac{21}{20}U]$ we apply (10). There $\sum_{p\le V} F(p)/p = 2\log\log V + O(1)$, and $\exp\{-\sum_{p\mid d} 2/(p-1)\} \le (\varphi(d)/d)^2$, because $\log(1-1/p) \ge -1/(p-1)$. So the inner sum over one box is $\ll U^{-1} \cdot U(\log U)\varphi(d)^{-1}(\varphi(d)/d)^2 = (\log U)\varphi(d)/d^2$. For $d = 1$ the same bound $\ll \log U$ follows from the elementary estimate for $q = 1$. Summing over $O(\log z)$ boxes, the inner sum is $\ll (\log z)^2 \varphi(d)/d^2$, uniformly in the residue. For the outer sum, partial summation from $\sum_{n\le V} F(n) \ll V \log V$ gives $\sum_{v\le z} F(v)/v \ll (\log z)^2$. Each pair therefore contributes $\ll (\log z)^4 \varphi(d)/d^2$. Finally, $\varphi(m)/m^2 \le 1/m$ and $[k,k'] = kk'/(k,k')$. Writing $k = ga$, $k' = gb$ with $g = (k,k')$ and then dropping $(a,b) = 1$,

$$\sum_{k,k'\le K} \frac{\varphi([k,k'])}{[k,k']^2} \le \sum_{g\le K} \frac{1}{g} \Big(\sum_{a\le K/g} \frac{1}{a} \Big)^2 \le \sum_{g\le K} \frac{(1+\log(K/g))^2}{g} \ll (1+\log K)^3. \quad \square$$

For $1 \in \mathcal{J} \subseteq \mathcal{K}(K)$ and $(c, L_{\mathcal{J}}) = 1$, and for a prime $\ell \in I_j$, let $f_{c;\mathcal{J}}(\ell)$ be the number of triples (k,u,v) with $k \in \mathcal{J}$, $(u,v) \in \mathcal{S}(k,c;H,z_j)$ and $4uv \mid k\ell + 1$.

Theorem 4.2 (supply)**.** *There are $a_h, A_h > 0$ and X_h, depending only on κ and the absolute analytic constants, such that for every $X \ge X_h$, every $1 \in \mathcal{J} \subseteq \mathcal{K}(K)$ and every c with $(c, L_{\mathcal{J}}) = 1$,*

$$a_h t^2 h(\mathcal{J}) \le \sum_{X^{1/2}<\ell\le X} \frac{f_{c;\mathcal{J}}(\ell)}{\ell} \le A_h t^2 h(\mathcal{J}). \tag{17}$$

Proof. Fix a block $(x, 2x]$ with $z = x^{1/6}$, so that $4uv \le 4x^{1/3}$. By (3), $\Lambda \ge \frac{1}{2}\log z = \frac{1}{12}\log x$. For $k \in \mathcal{J}$ and $(u,v) \in \mathcal{S}(k,c;H,z)$ we have $(k, 4uv) = 1$, and $4uv \mid k\ell + 1$ is the condition $\ell \equiv -k^{-1} \pmod{4uv}$, a reduced class. Hence

$$\sum_{x<\ell\le 2x} f_{c;\mathcal{J}}(\ell) = \sum_{k\in\mathcal{J}} \sum_{(u,v)\in\mathcal{S}(k,c;H,z)} \big[\pi(2x; 4uv, -k^{-1}) - \pi(x; 4uv, -k^{-1})\big].$$

Upper bound (for every block, the last one enlarged). By (8), $\varphi(4uv) \ge 2\varphi(u)\varphi(v)$ and (12), the sum is $\ll (x/\log x)\sum_{k\in\mathcal{J}} \varphi(k) k^{-2} (\log z)^2 \ll x \log x\, h(\mathcal{J})$. Divide by x and sum over the $O(t)$ blocks.

Lower bound (full blocks). Write each bracket as $(\mathrm{li}(2x)-\mathrm{li}(x))/\varphi(4uv)+E_x(4uv,-k^{-1})$. Since $\mathrm{li}(2x)-\mathrm{li}(x)\ge x/\log(2x)$ and $\varphi(4uv)\le 4uv$, (11) bounds the main terms below by

$$\frac{x}{\log 2x}\sum_{k\in\mathcal{J}}\sum_{\mathcal{S}(k,c;H,z)}\frac{1}{4uv}\ge\frac{x\Lambda^2 h(\mathcal{J})}{16\log 2x}\gg x\,t\,h(\mathcal{J}).$$

For the error terms, group the triples by $q=4uv$. The triples with a given (u,v) number $r_{\mathcal{J}}(u,v;c)$, and each triple with modulus q has error $|E_x(q,-k^{-1})|\le E_x^*(q)$. Put $W(q)=\sum_{4uv=q}r_{\mathcal{J}}(u,v;c)$, summed over coprime $H<u,v\le z$. There are at most $2^{\omega(q/4)}$ ordered coprime pairs with $uv=q/4$, so $W(q)^2\le 2^{\omega(q/4)}\sum_{4uv=q}r_{\mathcal{J}}(u,v;c)^2$ by Cauchy–Schwarz. By (8), $E_x^*(q)\ll x/(\varphi(q)\log x)$ for $q\le 4x^{1/3}$. Lemma 4.1 gives

$$\sum_q W(q)^2E_x^*(q)\ll\frac{x}{\log x}\sum_{\substack{H<u,v\le z\\(u,v)=1}}\frac{2^{\omega(uv)}r_{\mathcal{J}}(u,v;c)^2}{\varphi(4uv)}\ll x\,t^6,$$

using $\log z\asymp\log x\asymp t$ and $\log K\le\kappa t$. By Cauchy–Schwarz and (9) with $R=26$,

$$\Big|\sum_{\text{triples}}E_x\Big|\le\sum_q W(q)E_x^*(q)\le\Big(\sum_q W(q)^2E_x^*(q)\Big)^{1/2}\Big(\sum_{q\le 4x^{1/3}}E_x^*(q)\Big)^{1/2}\ll x\,t^{-10}.$$

Since $1\in\mathcal{J}$ gives $h(\mathcal{J})\ge 1$, this is $o(x\,t\,h(\mathcal{J}))$. Divide by $2x$ and sum over the $\asymp t$ full blocks. No constant depends on c, $\mathcal{J}$ or x. □

The Bombieri–Vinogradov input is used through the maximum $E_x^*(q)$ over all reduced classes. Therefore (17) holds for every choice of c and $\mathcal{J}$, including choices that depend on the data, with no averaging over c. The multiplicity $W(q)$ can be as large as a power of X. It is paid for in mean square by its own weight $1/\varphi(q)$, which is the role of Lemma 4.1.

Corollary 4.3 (fibre masses)**.** *Let $X\ge X_h$. For every residue c mod L_K, reduced or not, put*

$$\mathcal{J}_c=\{k\in\mathcal{K}(K):(k,c)=1\},\qquad\mu_c=\sum_{X^{1/2}<\ell\le X}\frac{f_c(\ell)}{\ell},$$

with f_c as in (7). Then $f_c=f_{c;\mathcal{J}_c}$, and

$$a_h t^2 h(\mathcal{J}_c)\le\mu_c\le A_h t^2 h(\mathcal{J}_c)\le C_u t^3 \tag{18}$$

with C_u depending only on κ.

Proof. An atom active in the fibre of c has $k\mid u+cv$ with $(u,k)=1$. Hence c is a unit modulo k, that is, $k\in\mathcal{J}_c$, and $(u,v)\in\mathcal{S}(k,c;H,z_j)$. Conversely, every triple counted by $f_{c;\mathcal{J}_c}(\ell)$ is an atom active in the fibre. Moreover $1\in\mathcal{J}_c$ and $(c,L_{\mathcal{J}_c})=1$, so Theorem 4.2 applies with $\mathcal{J}=\mathcal{J}_c$. Finally $h(\mathcal{J}_c)\le h(\mathcal{K}(K))\ll\log K\le\kappa t$ by Lemma 3.2. □

5. Factorial moments

For $y \geq 2$ let $S_y(n) = \mathbf{1}_{(n,P_y)=1} = \sum_{d \mid P_y} \mu(d)\mathbf{1}_{d \mid n}$, and write $(h)_m = h(h-1)\cdots(h-m+1)$.

Theorem 5.1 (upper factorial moments)**.** *For every $X \geq X_h$, every real $2 \leq y < K$ and every integer $m \geq 1$,*

$$\mathbb{E}\big((H_X)_m \,\big|\, S_y = 1\big) \leq (C_u t^3)^m. \tag{19}$$

Proof. The pair (c, S_y) is a function of $n \bmod \operatorname{lcm}(L_K, P_y)$. By Lemma 2.2, conditionally on c and $S_y = 1$ we can write $H_X = \sum_\ell I_\ell$, where the I_ℓ are independent and I_ℓ is Bernoulli with mean $p_\ell = f_c(\ell)/\ell$. For 0/1 variables, $(\sum_\ell I_\ell)_m$ is the sum of $\prod_{i \leq m} I_{\ell_i}$ over ordered m-tuples of distinct primes. Hence

$$\mathbb{E}\big((H_X)_m \,\big|\, c, S_y = 1\big) = \sum_{\ell_1,\dots,\ell_m \text{ distinct}} \prod_{i=1}^{m} p_{\ell_i} \leq \mu_c^m \leq (C_u t^3)^m$$

by Corollary 4.3, which holds for every c. Averaging over c gives (19). □

The base C_u does not depend on m; this matters because we shall take $m \asymp t^3$. The bound uses the upper estimate in *every* fibre, including fibres where c is not reduced modulo L_K. No Poisson approximation is involved.

6. An exponentially small conditioned void

The fibres c with $(c, L_K) > 1$ have a smaller multiplier set $\mathcal{J}_c$, and for them the lower bound in (18) can be weak. A small-prime selector makes such fibres exponentially rare. Put

$$y = Bt^3, \qquad S_y(n) = \mathbf{1}_{(n,P_y)=1}. \tag{20}$$

For fixed B and large X we have $2 \leq y < K$. Every non-representable integer without prime factors $\leq y$, in particular every non-representable prime $> y$, has $S_y = 1$.

Lemma 6.1 (conditioned void)**.** *There are $c_v > 0$ and B_0, depending only on κ, such that for every fixed $B \geq B_0$ and all $X \geq X_v(B)$,*

$$\mathbb{P}(H_X = 0 \mid S_y = 1) \leq \mathrm{e}^{-c_v t^3}. \tag{21}$$

Proof. Let

$$Z(c) = \sum_{\substack{y < p \leq K,\ p \mid L_K \\ p \mid c}} \frac{1}{p}.$$

The divisibility $p \mid c$ is meaningful because $p \mid L_K$. Conditionally on $S_y = 1$, the events $p \mid n$ for the primes $p > y$ with $p \mid L_K$ are independent with probabilities $1/p$, by the CRT, because S_y depends only on $n \bmod P_y$. Since $0 < y/p < 1$ and $\mathrm{e}^s - 1 \leq (\mathrm{e}-1)s$ on $[0,1]$,

$$\mathbb{E}(\mathrm{e}^{yZ} \mid S_y = 1) = \prod_{\substack{y < p \leq K \\ p \mid L_K}} \Big(1 + \frac{\mathrm{e}^{y/p} - 1}{p}\Big) \leq \exp\Big\{(\mathrm{e}-1)y \sum_{n > y} n^{-2}\Big\} \leq \mathrm{e}^4,$$

using $\sum_{n>y} n^{-2} \leq 1/\lfloor y \rfloor \leq 2/y$. Therefore $\mathbb{P}(Z > \eta \mid S_y = 1) \leq \mathrm{e}^{4-\eta y}$ for every $\eta > 0$.

By (14) there are absolute $a_0 > 0$ and K_1 with $h(\mathcal{K}(K)) \geq a_0 \log K$ for $K \geq K_1$. On $S_y = 1$ no prime $p \leq y$ divides c. A multiplier $k \notin \mathcal{J}_c$ therefore has a prime factor p with $y < p \leq K$, $p \mid L_K$ and $p \mid c$. By (15),

$$h(\mathcal{J}_c) \geq h(\mathcal{K}(K)) - \sum_{\substack{y<p\leq K,\ p\mid L_K \\ p\mid c}} \frac{1+\log K}{p} \geq (a_0 - 2Z(c)) \log K.$$

Put $\eta = a_0/4$. On the event $Z(c) \leq \eta$ we get $h(\mathcal{J}_c) \geq \frac{a_0}{2} \log K \geq \frac{a_0 \kappa}{4} t$. By Lemma 2.2 and Corollary 4.3,

$$\mathbb{P}(H_X = 0 \mid c, S_y = 1) = \prod_{\ell} \Big(1 - \frac{f_c(\ell)}{\ell}\Big) \leq \mathrm{e}^{-\mu_c} \leq \mathrm{e}^{-a_h t^2 h(\mathcal{J}_c)} \leq \mathrm{e}^{-a_v t^3},$$

where $a_v = a_h a_0 \kappa/4$. Choose $B_0 = 2a_v/\eta$. For $B \geq B_0$, averaging over c gives $\mathbb{P}(H_X = 0 \mid S_y = 1) \leq \mathrm{e}^{-a_v t^3} + \mathrm{e}^{4-2a_v t^3} \leq \mathrm{e}^{-a_v t^3/2}$ for large X. Take $c_v = a_v/2$. □

A Markov bound for Z would be far too weak here. The exponential moment at parameter $y \asymp t^3$ makes the bad fibres as rare as the required saving. The price is the expansion of S_y into $2^{\pi(y)}$ classes, which is affordable. Revealing c is an operation inside the CRT computation only. It does not expand anything into the L_K progressions modulo L_K.

7. Bonferroni assembly on the integers

Lemma 7.1 (even Bonferroni majorant)**.** *For an even integer $r \geq 0$ and an integer $h \geq 0$ let $Q_r(h) = \sum_{j=0}^{r} (-1)^j \binom{h}{j}$. Then $Q_r(0) = 1$, $Q_r(h) = \binom{h-1}{r} \geq 0$ for $h \geq 1$, and*

$$\mathbf{1}_{h=0} \leq Q_r(h) \leq \mathbf{1}_{h=0} + \binom{h}{r+1}. \tag{22}$$

Proof. For $h \geq 1$, Pascal's rule telescopes the sum to $(-1)^r \binom{h-1}{r}$. This vanishes for $h \leq r$, and for $h \geq r+1$ it is at most $\frac{h}{r+1}\binom{h-1}{r} = \binom{h}{r+1}$. □

Theorem 7.2 (assembly)**.** *Let $B \geq B_0$ be fixed and $y = Bt^3$. There are constants $c_a, D_B, C_L > 0$ and a threshold X_a, depending only on κ and B, with the following property. For $X \geq X_a$, let r be the least even integer $\geq D_B t^3$ and put*

$$\nu_X(n) = S_y(n)\, Q_r(H_X(n)). \tag{23}$$

Then $\nu_X \geq 0$ on $\mathbb{Z}$, and $\nu_X(n) \geq 1$ for every non-representable n with $(n, P_y) = 1$. Moreover $\mathbb{E}\nu_X \leq \mathrm{e}^{-c_a t^3}$, and for every real $N \geq 1$

$$\sum_{1\leq n\leq N} \nu_X(n) \leq N\mathrm{e}^{-c_a t^3} + \mathrm{e}^{C_L t^4}. \tag{24}$$

Proof. Mean. By (22), Lemma 6.1 and Theorem 5.1, using $m! \geq (m/\mathrm{e})^m$,

$$\mathbb{E}(Q_r(H_X) \mid S_y = 1) \leq \mathrm{e}^{-c_v t^3} + \frac{\mathbb{E}((H_X)_{r+1} \mid S_y = 1)}{(r+1)!} \leq \mathrm{e}^{-c_v t^3} + \Big(\frac{\mathrm{e}C_u t^3}{r+1}\Big)^{r+1}.$$

Choose $D_B > \mathrm{e}C_u$ with $D_B \log(D_B/(\mathrm{e}C_u)) \geq 2c_v$. The function $m \mapsto m \log(m/(\mathrm{e}C_u t^3))$ is increasing for $m \geq C_u t^3$, and $r + 1 \geq D_B t^3$, so the last term is at most $\mathrm{e}^{-2c_v t^3}$. Multiplying by $\mathbb{P}(S_y = 1) \leq 1$ gives $\mathbb{E}\nu_X \leq \mathrm{e}^{-c_a t^3}$ with $c_a = c_v/2$ for large X. The nonnegativity and the lower bound for non-representable n follow from Lemma 7.1 and $H_X(n) = 0$ (Section 2).

Expansion into classes. For every integer n,

$$\binom{H_X(n)}{j} = \sum_{\substack{\mathcal{B}\subseteq\mathcal{A}_X\\ |\mathcal{B}|=j}} \prod_{A\in\mathcal{B}} \mathbf{1}_{E_A}(n),$$

so that

$$\nu_X(n) = \sum_{d|P_y} \sum_{j=0}^{r} \sum_{|\mathcal{B}|=j} (-1)^j \mu(d)\, \mathbf{1}_{d|n} \prod_{A\in\mathcal{B}} \mathbf{1}_{E_A}(n).$$

Each product is the indicator of the empty set or of a single class modulo $q = \operatorname{lcm}(d, k_A\ell_A : A \in \mathcal{B})$, and q divides $\mathcal{M}$. Hence $\mathbb{E}\nu_X$ is exactly the sum of the coefficients divided by these moduli. The number of terms, which bounds the absolute coefficient sum, is

$$T_{\mathrm{abs}} \le 2^{\pi(y)} \sum_{j=0}^{r} \binom{|\mathcal{A}_X|}{j} \le 2^{\pi(y)}(r+1)|\mathcal{A}_X|^r.$$

For a crude inventory, an atom is determined by $k \le K$, $\ell \le X$ and $u, v \le X^{1/6}$, so $|\mathcal{A}_X| \le KX^{4/3} \le \mathrm{e}^{(4/3+\kappa)t}$. With $r \le D_B t^3 + 2$ and $\pi(y) \le y = Bt^3$,

$$\log T_{\mathrm{abs}} \le Bt^3 \log 2 + \log(r+1) + r(4/3+\kappa)t \le C_L t^4 \tag{25}$$

for a suitable $C_L = C_L(\kappa, B)$ and large X. The exponent is $O(rt)$ and not $O(r)$: every factor $\binom{\cdot}{j}$ costs a choice of atoms among $\mathrm{e}^{O(t)}$.

Counting. For every modulus $q \ge 1$ and every class $a \bmod q$,

$$\#\{1 \le n \le N : n \equiv a\ (q)\} = N/q + \theta, \qquad |\theta| \le 1.$$

This also holds when $q > N$. Summing over the terms with their signed coefficients gives $\sum_{n\le N} \nu_X(n) = N\mathbb{E}\nu_X + O^*(T_{\mathrm{abs}})$, where $|O^*(T)| \le T$. This proves (24). □

No prime-counting estimate enters at scale N, and the residue classes modulo L_K are never enumerated: the term moduli are at most $P_y(KX)^r = \mathrm{e}^{O(t^4)}$, while L_K itself plays a role only in the computation of $\mathbb{E}\nu_X$.

8. Proof of Theorem 1.1

Primes. Let N be large, $L = \log N$, and take $t = \alpha L^{1/4}$ and $X = \mathrm{e}^t$, where $\alpha > 0$ is fixed with $C_L\alpha^4 \le 1/2$. Every non-representable prime $p > y$ satisfies $\nu_X(p) \ge 1$. By (24),

$$E_{\mathrm{pr}}(N) \le y + N\mathrm{e}^{-c_a\alpha^3 L^{3/4}} + \mathrm{e}^{C_L\alpha^4 L} \le B\alpha^3 L^{3/4} + N\mathrm{e}^{-c_a\alpha^3 L^{3/4}} + N^{1/2}.$$

This is $\ll N\exp\{-c_a\alpha^3 L^{3/4}\}$. Enlarging the constant covers bounded $N \ge 3$. □

All denominators. If $p \mid n$ and $4/p = \sum_i 1/x_i$, then $4/n = \sum_i 1/((n/p)x_i)$. Hence every prime factor of a non-representable n is non-representable, and the non-representable integers lie in the multiplicative semigroup generated by the non-representable primes (together with 1).

By the prime case there are $c_0 > 0$ and $x_0 \ge \mathrm{e}$ with $E_{\mathrm{pr}}(x) \le x\mathrm{e}^{-c_0(\log x)^{3/4}}$ for $x \ge x_0$. Fix $0 < \eta_0 < c_0$, put $L = \log x$ and $\delta = \eta_0 L^{-1/4}$, and assume x is so large that $\delta < 1/4$. Rankin's trick gives

$$E(x) \le \sum_{n \text{ non-repr.}} \left(\frac{x}{n}\right)^{1-\delta} \le x^{1-\delta} \prod_{\substack{p\le x\\ p \text{ non-repr.}}} (1 - p^{-1+\delta})^{-1}. \tag{26}$$

The prime powers p^a with $a \ge 2$ contribute $O(\sum_p p^{-3/2}) = O(1)$ to the logarithm of the product, and the primes below x_0 contribute $O_{x_0}(1)$. By partial summation,

$$\sum_{\substack{x_0 < p \le x \\ p \text{ non-repr.}}} p^{-1+\delta} \le \mathrm{e}^{\delta L - c_0 L^{3/4}} + \int_{\log x_0}^{L} \mathrm{e}^{\delta s - c_0 s^{3/4}}\, ds.$$

For $s \le L$ we have $\delta s = \eta_0 s L^{-1/4} \le \eta_0 s^{3/4}$. So the first term is at most 1, and the integral is at most $\int_0^\infty \mathrm{e}^{-(c_0-\eta_0)s^{3/4}} ds < \infty$. The product in (26) is therefore bounded uniformly in x, and $E(x) \ll x^{1-\delta} = x \exp\{-\eta_0 (\log x)^{3/4}\}$. □

9. Remarks

Remark 9.1 (comparison with Vaughan's argument). Vaughan uses the classes of Lemma 2.1 with $k = 1$ and Montgomery's large sieve with prime moduli (we follow the account in [PW], [LL]; the following balance is heuristic). With $k = 1$ only, the mass in (17) is $\asymp t^2$. The large sieve gains $\exp\{c\mu\}$ only at levels Q with $\log Q \gtrsim \mu \log X$. Since $Q \le N^{1/2}$, this forces $t^3 \lesssim \log N$ and gives $(\log N)^{2/3}$. Here the extra factor $h(\mathcal{K}(K)) \asymp \log K \asymp t$ in the mass is paid for by the same kind of budget: the Bonferroni degree is $r \asymp \mu$ and each degree costs a factor $\mathrm{e}^{O(t)}$ in (25). The balance becomes $t^4 \asymp \log N$. Using the multipliers in the large sieve directly, as in [LL], would require splitting $[1, N]$ modulo L_K, which is impossible once $\log L_K \asymp K$ exceeds $\log N$. This is why [LL] is limited to $K \ll \log N$ and to a gain of $(\log \log N)^{1/3}$.

Remark 9.2 (total identity-class supply). For $M \equiv 3 \pmod 4$ put $A = (M+1)/4$ and $\mathcal{R}(M) = \{-uv^{-1} \bmod M : uvw = A\}$, the set of classes forced by Lemma 2.1 with $k\ell$ replaced by M. Then $\mathcal{R}(M) = \{-4D \bmod M : D \mid A^2\}$. Indeed $-uv^{-1} \equiv -4u^2 w$ because $4uvw \equiv 1$, so one may take $D = u^2 w$. Conversely, if $p^e \| A$ and $p^d \| D$, give u, w, v the exponents $\lfloor d/2 \rfloor$, $d - 2\lfloor d/2 \rfloor$ and the remainder. Divisors $D < A$ give distinct classes, since $0 < 4D < M$, so $\frac{1}{2}(\tau(A^2) - 1) \le |\mathcal{R}(M)| \le \tau(A^2)$. From $\tau(n^2) = \sum_{d^2 \mid n} \mu(d) \tau_3(n/d^2)$ and $\sum_{m \le x} \tau_3(m) \sim \frac{1}{2} x \log^2 x$ one gets $\sum_{M \le Q,\, M \equiv 3\,(4)} |\mathcal{R}(M)|/M \asymp (\log Q)^3$. So the supply of all identity classes, over all moduli and with distinct residues, has exponent three, and the multipliers of this note reach that order of magnitude. Combined with the heuristic balance above, this explains the value 3/4. The rigorous statement that 3/4 cannot be beaten by majorants of the type used here is the internal result quoted in Remark 1.2.

Remark 9.3 (numerator m). The argument transfers to m/n for fixed $m \ge 4$. The identity becomes $k\ell + 1 = m\,uvw$ with $(k, m) = 1$, and the moduli become muv. This gives $E_m(N) \ll N \exp\{-c(\log N)^{3/4} m^{-1/4}\}$ with absolute c. That extension is recorded in internal working notes and is not part of the claims of this note.

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